



Zongmin Zhang

HKUST, BEng Computer Science | Minor Chemistry

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🌐 Website 📄 Scholar 🆔 ORCID 🐙 GitHub 🔗 LinkedIn

EDUCATION

The Hong Kong University of Science and Technology (HKUST) 09/2023 – present

BEng Computer Science, School of Engineering; Minor in Chemistry

CGA: 3.999/4.300 | Dean's List ×4 | HKSAR Government Scholarship ×2

Selected A+ coursework: PHYS1007 Quantum Information for Everyone, CHEM1020 General Chemistry I, CHEM1030 General Chemistry II, CHEM2111 Fundamentals of Organic Chemistry, COMP2011 Programming with C++, COMP2012 Object-Oriented Programming and Data Structures, COMP3021 Java Programming, COMP3711 Design and Analysis of Algorithms, COMP4451 Game Programming, MATH2011 Introduction to Multivariable Calculus, MATH2111 Matrix Algebra and Applications, MATH2411 Applied Statistics, MATH2421 Probability

PUBLICATIONS & PREPRINTS

- [1] **Zongmin Zhang**, Yuyang Lou, Bowen Zhang, Junwu Chen, Ryo Kuroki, Xuan Vu Nguyen, Edwin Fako, Lixue Cheng, Philippe Schwaller. “AdsMind: A Physics-Grounded Multi-Agent System for Self-Correcting Discovery of Adsorption Configurations on Heterogeneous Catalyst Surfaces.” *Under review; arXiv preprint [arXiv:2606.19152](https://arxiv.org/abs/2606.19152)*, 2026.
- [2] Zhilong Song, **Zongmin Zhang**, Lixue Cheng. “Autonomous Heterogeneous Catalyst Discovery with a Self-Evolving Multi-Agent Digital Twin.” *Under review; arXiv preprint [arXiv:2606.05050](https://arxiv.org/abs/2606.05050)*, 2026.
- [3] Aritra Roy et al. (incl. **Zongmin Zhang**). “From Knowledge to Action: Outcomes of the 2025 LLM Hackathon for Applications in Materials Science and Chemistry.” *arXiv preprint [arXiv:2605.03205](https://arxiv.org/abs/2605.03205)*, 2026.

RESEARCH EXPERIENCE

Final Year Project, HKUST 2026 – 2027 (expected)

Advisors: Prof. Chaojian Li & Prof. Lixue Cheng

Embodied Multi-Agent System for High-Stakes Environment Safety: Transferring Autonomous Driving Architectures to Edge-Deployed Labs

- Project aims to combine edge-deployed multi-agent reasoning, multimodal sensor fusion, and hardware-in-the-loop (HIL) control for laboratory safety.
- Planned evaluation: sandbox tests of local fallback, safety interrupts, and end-to-end latency.

UGRA, AI4PhysSci Lab, HKUST 02/2026 – present

Supervisor: Prof. Lixue Cheng

- Leading the development of AdsMind, a physics-grounded multi-agent system for autonomous adsorption-configuration discovery on heterogeneous catalyst surfaces. Designed the closed-loop architecture integrating generative agent proposals with DFT-based physics validation and self-correction.
- Built the benchmark framework and conducted cross-backend evaluation (Codex, Claude Code) across diverse catalyst surfaces, establishing the first systematic multi-agent benchmark for adsorption structure prediction.
- Contributed to CatDT, a self-evolving multi-agent digital-twin framework for autonomous heterogeneous catalyst discovery. Designed and executed Codex-based ablation experiments to isolate agent capability contributions.

Project Student, LIAC, ISIC, EPFL 09/2025 – 01/2026

Supervisors: Prof. Philippe Schwaller & Dr. Edwin Fako

- Developed agentic simulation methods for atomic rearrangement and surface reactivity prediction on catalyst surfaces, combining language-model reasoning with atomistic simulation backends.
- This semester project established the EPFL–HKUST collaboration that produced the AdsMind system.

UROP, HKUST (Prof. Xiaojuan Ma) 02/2025 – 05/2025

User Experience & Visual Representation in Virtual Reality

UROP, HKUST (Prof. Raymond Chi-Wing Wong) 06/2024 – 12/2024

Knowledge Discovery Over Database

HONORS & SCHOLARSHIPS

Dean's List (4 consecutive semesters: Fall 2023 – Spring 2026)	02/2024 – 09/2026
HKSAR Government Scholarship (2024/25, 2025/26; HK\$80,000 each)	12/2024, 12/2025
HK, China–APEC Scholarship (2024/25)	12/2024
HKUST Study Abroad Grant & Reaching Out Award (HK\$10,000)	12/2025

OVERSEAS & INDUSTRY EXPERIENCE

Exchange Student, EPFL, Switzerland <i>School of Computer and Communication Science</i>	09/2025 – 02/2026
APRU Virtual Exchange, SJTU <i>Coursework: MSE2602 Materials Chemistry</i>	02/2025 – 06/2025
Computer Vision Algorithm Intern <i>Sansen Well (Shanghai) Robotics Co., Ltd., Shanghai</i>	06/2025 – 08/2025
• Independently built an end-to-end multi-view 3D reconstruction pipeline for industrial wire-rope inspection.	

TEACHING & SERVICES

Reviewer: <i>MLST</i> (IOP), NeurIPS'26 AI4S Workshop, ICML'26 AI4S Workshop	2026
UGTA: COMP2011 (Sp25), CSE Programming Commons (Fa24, Sp26)	2024 – 2026
Student Representative, SENG Mainland UG Recruitment, Guangzhou	06/2026 – 07/2026
Founding Organizer, HKUST AI for Chemistry Club (DAG in progress)	2026 – present

ADDITIONAL

Technical skills: Python, C/C++, Java (proficient); Scala, C#, JavaScript, Rust. PyTorch, NumPy, TensorRT. LangChain, LangGraph, MCP, RAG. RDKit, ASE. YOLO, SAM, OpenCV, 3D reconstruction. Linux, shell scripting. Godot, Unity. Streamlit, VTK, Meta XR Toolkit.

Languages: Mandarin (native); English (professional; Gaokao 142/150); Japanese (JLPT N3); French (CEFR A1); Cantonese (basic).

Other awards: Finalist, HKUST GenAI Hackathon (03/2026); First Prize, 5th National Data Analysis Competition, Theory Track (2026).